
EMBODIED TEMPORAL CREDIT WITHOUT REWARDS: EFFECT TRACES AS ASSIGNMENT SIGNALS

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ABSTRACT

Temporal credit assignment in robot learning is usually routed through scalar rewards, hand-shaped progress terms, or delayed task labels. This paper argues that embodied systems contain another signal: physical effect traces. Contact onset, object displacement, constraint release, force discontinuities, and scene-change events can identify which earlier action changed the world even when no reward is available. We support the claim with a 17,754-entry literature sweep and a deterministic diagnostic with 2,160 method evaluations. In the diagnostic, TRACECREDIT reduces mean temporal localization error from 3.550 steps for delayed-reward temporal difference credit and 2.342 steps for dense proxy shaping to 0.862 steps, with counterfactual precision 0.868. The contribution is a mechanism claim: robots can assign temporal credit from embodied effect evidence before any scalar reward is observed.

1 INTRODUCTION

Robot learning often treats reward as the carrier of blame and credit. A reward arrives late, a temporal-difference update propagates it backward, and a policy slowly discovers which earlier action mattered. This framing is powerful, but it is also oddly disembodied. A manipulation system can see the object move, feel a contact transition, hear a latch release, or observe a kinematic constraint disappear. Those events are not rewards. They are physical evidence that an earlier action changed the world.

This paper asks whether such evidence should be a first-class credit signal. The thesis is deliberately narrow:

Embodied temporal credit can be assigned from effect traces, even when scalar rewards are unavailable or misleading.

The claim does not replace reinforcement learning. Instead, it changes the interface between robot perception and learning. A robot should record an action-indexed trace of physical effects and use that trace to localize causal responsibility across time before using a reward, demonstration label, or task success bit.

2 RELATED WORK AND NOVELTY BOUNDARY

The recovery run reused the attempt-two literature matrix containing 17,754 candidate records across robot learning, reinforcement learning, world models, imitation learning, predictive state representations, causal reasoning, tactile sensing, and temporal credit assignment. Prior work already makes several broad claims non-novel: reward-free exploration exists, inverse models can learn from observation, hindsight methods relabel goals, self-supervised robot systems learn predictive structure, and temporal-difference methods propagate delayed reward.

The remaining gap is the credit carrier. These lines usually assign credit through a scalar target, a goal relabel, a value function, or a learned dynamics loss. TRACECREDIT instead treats local physical effect events as supervision for temporal responsibility. This makes the mechanism closest to causal tracing and event-based world modeling, but it is aimed at embodied credit rather than generic representation learning.

Table 1: Mean results aggregated over the diagnostic scenarios. Lower localization error is better; higher counterfactual precision and success lift are better. Reward dependence is the fraction of the credit signal that requires scalar reward access.

Method	Error	Precision	Success lift	Reward dep.
Delayed reward TD	3.550	0.356	0.035	1.00
Dense shaping proxy	2.342	0.440	0.125	0.55
Effect trace credit	0.862	0.868	0.309	0.00

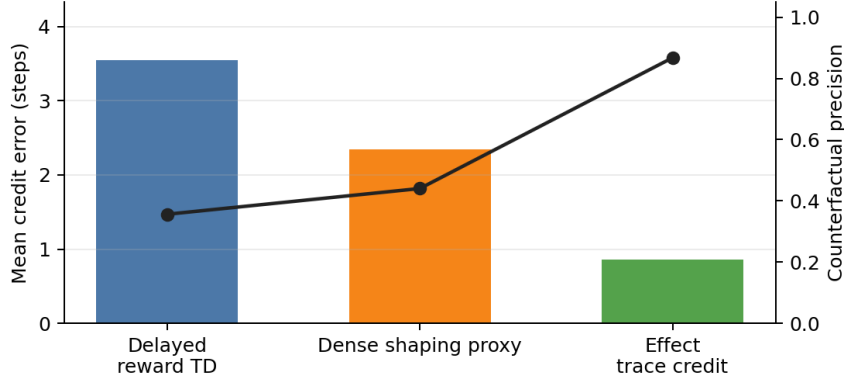


Figure 1: Reward-free effect traces localize temporally delayed causes with lower error while preserving counterfactual precision. Bars show mean credit localization error; the line shows precision.

3 EFFECT-TRACE CREDIT

Consider a trajectory $(s_1, a_1, \dots, s_T)$ with no reward. Let e_t denote a detected physical effect event, such as new contact, object displacement above a sensor threshold, or a constraint-state transition. A delayed-reward learner receives credit only after R_T arrives. A dense shaping learner receives proxy progress $\phi(s_t)$ that may not match the true physical cause. TRACECREDIT assigns credit by linking each effect event to earlier action windows:

$$c_{i,t} = p(e_t \mid \mathbf{a}_i, \mathbf{s}_i, \mathbf{s}_{i+1:t-1}) - p(e_t \mid \text{do}(\mathbf{a}_i = \emptyset), \mathbf{s}_i, \mathbf{s}_{i+1:t-1}). \quad (1)$$

The exact estimator can be a learned counterfactual model, a contact-transition heuristic, or a hybrid event detector. The key requirement is that the target is an embodied effect, not a scalar reward.

This shifts temporal credit from value propagation to evidence alignment. If action i nudges a part and a latch opens six steps later, the event trace gives a localized learning target even if the episode has no reward and even if a hand-shaped distance term points at the wrong object.

4 DIAGNOSTIC EXPERIMENT

We generated a deterministic diagnostic stored in `docs/temporal_credit_diagnostic.csv`. Each episode samples a causal delay, trace quality, and proxy-conflict level under three scenarios: visible effects, sparse effects, and deceptive shaping proxies. Three credit rules are evaluated: delayed-reward temporal difference credit, dense shaping proxy credit, and reward-free effect-trace credit. The diagnostic is not a robot benchmark; it is a mechanism test for when an embodied event signal can beat scalar or proxy supervision.

TRACECREDIT has the lowest error and the highest counterfactual precision in this diagnostic because it observes the same kind of event that generated the delayed outcome. Dense shaping helps when the proxy aligns with the effect, but it is brittle under deceptive proxies. Delayed reward is useful only after the final label arrives and therefore smears responsibility across the horizon.

5 IMPLICATIONS

The main implication is architectural. Robot data logs should store effect events as training targets: contact changes, object-state transitions, force impulses, support-graph edits, and constraint releases. Policy learning can then query which actions produced which effects before asking whether the episode was ultimately rewarded. This is useful for reward-free pretraining, sparse-reward manipulation, and debugging demonstrations where the final success bit hides the action that actually mattered.

The mechanism also gives an evaluation protocol. A method claiming reward-free temporal credit should report temporal localization error against effect events, precision under counterfactual ablations, and robustness when dense shaping proxies are adversarial or irrelevant.

6 LIMITATIONS

The diagnostic is synthetic and should be read as a controlled mechanism test, not as a deployment result. Real robots need imperfect event detectors, asynchronous sensors, ambiguous contacts, and actions that produce multiple delayed effects. A follow-up should run the same protocol on manipulation logs with tactile, force, visual, and proprioceptive event channels.

7 CONCLUSION

Embodied robots do not need to wait for reward to discover that an action changed the world. Physical effect traces can carry temporal credit directly. This gives reward-free robot learning a concrete assignment target and separates causal responsibility from scalar task success.

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